Qiang Ma

Multimodal learning / Efficient vision-language models
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Computer Science PhD student at HKU focused on multimodal learning and efficient vision-language models. Research spans 3D medical image classification, uncertainty-aware fusion, and reinforcement learning for efficient reasoning.

Research Experience

The University of Hong Kong 09/2025 - 08/2026
Research Assistant | Advisor: Lequan Yu Hong Kong

- **Efficient 3D vision-language models:** Proposed PRISM, combining spatio-anatomical evidence scoring and foveated token routing to retain diagnostic regions and compress redundant context.
- **Efficiency and evaluation:** Achieved strong classification performance using only **15.6%–18.8% of visual tokens**; evaluated the efficiency-performance trade-off on bone tumor CT and cardiac MRI benchmarks.

West China Biomedical Big Data Center, West China Hospital 08/2024 - 08/2025
Research Intern | Advisor: Huan Song Chengdu, China

- **Robust multimodal fusion:** Proposed AURA for cross-center medical prediction with missing modalities at inference. Quantified epistemic uncertainty and used uncertainty-guided routing and calibration to improve reliability.
- **Cross-center validation:** Evaluated on **five datasets across the US, China, and Korea**, with superior performance in cross-center transfer and missing-modality settings.

Education

The University of Hong Kong 09/2026 - 06/2030 (expected)
PhD Student in Computer Science | Advisor: Lequan Yu

University of Electronic Science and Technology of China 09/2023 - 06/2026
MS in Computer Science | Advisor: Xiaoshuang Shi

Harbin Engineering University 09/2018 - 06/2022
BS in Computer Science

Publications and Manuscripts

* Co-first authors

- **PRISM: Optimizing 3D Medical VLLMs for Classification via Spatio-Anatomical Evidence Purification**
  Qiang Ma, Feng Wu, Huahui Yi, et al.
  Under Review.
- **Shapley-Inspired Causal Credit Assignment for Reinforcement Learning in Efficient Reasoning**
  Ziyuan Qin*, **Qiang Ma***, Haoyu Wang, Huahui Yi, Lequan Yu.
  Under Review.
- **NeuroCogMap Reveals Cognitive Organization of Large Language Models**
  Zhongxiang Sun*, Haolang Lu*, **Qiang Ma**, et al.
  Nature, Under Review, 2026.
- **Robust Multimodal Fusion via Epistemic Uncertainty Quantification for Cross-Center Medical Prediction**
  Qiang Ma, Ziyuan Qin, He Lyu, et al.
  Information Fusion, Under Review.
- **Interpretable Multi-View Fusion Network for Alzheimer's Disease Diagnosis with Large-scale Pre-trained Vision-Language Model**
  Qiang Ma, Jinghao Xu, Tengfei Li, et al.
  IEEE BIBM, 2025.
- **Diagnostic Evidence-Based Prototype Learning via Dirichlet Process for Multimodal Bone Tumor Subtyping**
  Feng Wu, Ruiling She, Ruiyang Zhang, Liang Peng, **Qiang Ma**, Yinghua Zhao, Lequan Yu.
  MICCAI, 2026.

Research Expertise and Service

Methods: Multimodal fusion; vision-language models; token routing; epistemic uncertainty estimation and calibration; reinforcement learning for efficient reasoning.

Reviewer: WACV 2027; MICCAI FLARE Challenge 2026; ISBI 2026; Pattern Recognition.

Teaching Assistant: COMP2501 Introduction to Data Science, The University of Hong Kong.